

WatchMe

Computer Vision Speech Practice Agent

Final Project Capstone · Raymond Newton · ITAI 1378

LEARNER
READY

MULTIM
ODAL
AGENT

HUMAN
OVERSIGHT



PERCEIVE → REASON →
TEACH

Problem & Motivation

A multimodal prototype for learner readiness and speech practice

The learning problem

Speech practice needs clear prompts, timing, and feedback.
Camera and microphone conditions can affect learner interaction.
Raw video/audio alone is hard to inspect or reproduce.
Automated feedback must stay limited and reviewable.



watchMe goal

Use computer vision to check learner readiness.
Track mouth visibility and movement during practice.
Run short CVC and Shapes lessons with typed or microphone input.
Save traces so results, errors, and limitations can be reviewed.

Project Objectives

What the final capstone system proves

1 Ingest

Accept images, folders, webcam frames, typed input, and microphone input.

2 Perceive

Detect face count, position, distance, mouth visibility, and mouth movement.

3 Reason

Convert perception into READY, REPOSITION, MOVE_CLOSER, or other actions.

4 Teach

Start a lesson, display prompts, record speech, and score responses.

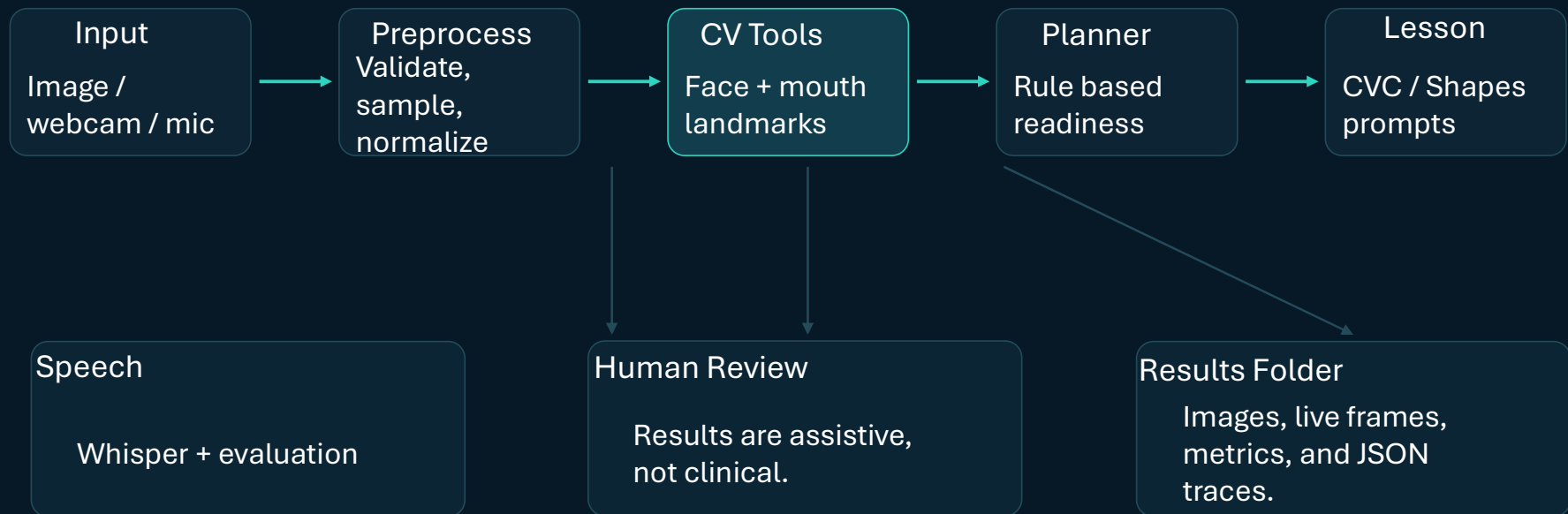
5 Document

Package code, tests, traces, failures, ethics, and AI usage for GitHub review.

Project success means a complete, explainable prototype, not clinical speech diagnosis.

System Architecture

Camera + audio → perception → planner → lesson → trace



Computer Vision Components

Face readiness, mouth analysis, and temporal movement

Face Perception

MediaPipe Face Landmarker detects one learner, face count, bounding box, centering, and face-size ratio.



Mouth Analysis

Landmarks estimate mouth visibility, open/closed state, and mouth-open ratio for speech readiness.



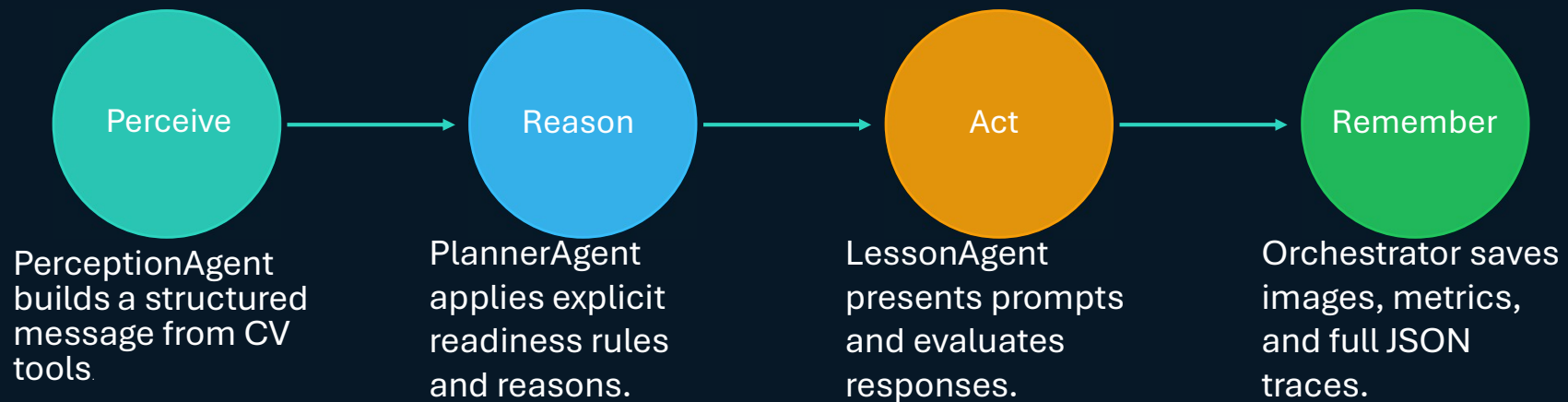
Temporal Movement

Webcam frames are compared over time to detect visible mouth movement during the lesson.



Agent Workflow

Perception → reasoning → lesson action → trace



The agent assists practice. It does not diagnose speech ability or make high-stakes decisions.

Implementation & Repository

A complete GitHub-ready project structure

```
watchMe/  
├── README.md  
├── requirements.txt  
├── agents/  
├── tools/  
├── lessons/  
├── data/  
├── models/  
├── results/  
└── docs/  
    ├── architecture.md  
    ├── test_plan.md  
    ├── ethics.md  
    └── AI_usage_log.md
```

Core technologies

Python
OpenCV
MediaPipe Face Landmarker
Whisper tiny.en
SoundDevice / SoundFile
Structured JSON traces
GitHub documentation

Testing Methodology

How the system was validated before submission

- 1 Import test Core agents import cleanly from a fresh command.
- 2 Visual test Frozen 14-scenario evaluation passed 14/14.
- 3 Robustness test Dark, blur, rotation, blank, and corrupt inputs handled safely.
- 4 Live test Camera, planner, lesson, microphone, Whisper, and trace ran together.
- 5 Submission test README, results, docs, GitHub repo, and reproducibility path verified.

Failures were kept as evidence: no-input runs, speech errors, planner conflicts, and API mismatches.

Results & Outputs

Final measurable evidence from watchMe

14/14

visual scenarios
passed

2/3

final spoken
prompts correct

0

final no-input
events

1

public GitHub repo

Final validation snapshot

Session live_lesson_20260809_212807 completed.
731 frames, 48 perception updates, 1 face detected.
Automatic countdown started the Shapes lesson.
Mouth movement detected with 18 events and max delta 0.2841.
Circle and square were correct; triangle was a Whisper transcription error.

Limitations, Risks & Responsible AI

Useful assistive feedback without overclaiming

Speech limits

Background noise, mic placement, and short words can cause Whisper errors.

Vision limits

Lighting, rotation, distance, and camera angle can affect readiness decisions.

Privacy

Camera frames, face images, audio, and traces can contain identifiable data.

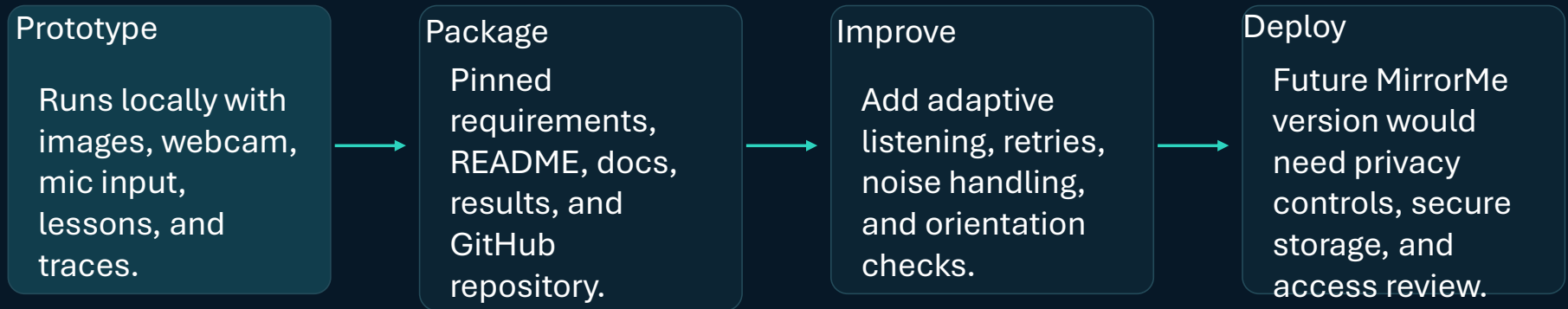
Bias and context

Performance may vary across people, cameras, lighting, speech patterns, and hardware.

Guardrail: watchMe supports practice and human review. It is not a clinical or diagnostic system.

Deployment Plan & Future Work

From capstone prototype to stronger MirrorMe module



Next improvements

- Longer or adaptive speech listening window.
- Limited retry before moving to the next prompt.
- Optional typed fallback after failed speech attempts.
- ProgressAgent with local SQLite session history.
- More diverse camera and speech evaluation data.

Conclusion

watchMe demonstrates a complete multimodal CV agent workflow

What this capstone shows

- Face and mouth computer vision
- Agent-style perception and planning
- Live learner-facing lesson interface
- Speech input with Whisper
- Traceable testing and failure analysis
- Responsible AI documentation

Submission links

- GitHub: <https://github.com/ydyllc/watchMe-ITAI1378-Capstone>
- Demo video: optional / to be added if submitted
- Slides:
WatchMe_Final_Project_Capstone_ITAI1378_Final.pptx

Final takeaway: the goal is not a perfect speech score, but a system that can perceive, reason, teach, log results, and honestly explain its limits.